SAVLG-CBM Ablation Methodology

April 10, 2026

1 Pipeline

For an input image x, the SAVLG model produces a global concept branch and a spatial concept branch.

1.1 Backbone Features

Let the frozen visual backbone produce:

$$\mathbf{f}_4 \in \mathbb{R}^{C_4 \times H_4 \times W_4}, \qquad \mathbf{f}_5 \in \mathbb{R}^{C_5 \times H_5 \times W_5}.$$

For the ResNet-50 MMPretrain CUB setup with 448×448 inputs:

$$H_4 \times W_4 = 28 \times 28, \qquad H_5 \times W_5 = 14 \times 14.$$

1.2 Global Branch

The global branch uses the conv5 feature map. Global average pooling gives

$$\bar{\mathbf{f}}_5 = \mathrm{GAP}(\mathbf{f}_5),$$

and the global concept head produces

$$\mathbf{c}_{\text{global}} = g(\bar{\mathbf{f}}_5) \in \mathbb{R}^K,$$

where K is the number of concepts.

1.3 Spatial Branch

The spatial branch predicts concept maps

$$\mathbf{M} \in \mathbb{R}^{K \times H \times W}.$$

The exact branch depends on the ablation:

conv5 branch.

$$\mathbf{M} = s(\mathbf{f}_5), \qquad H \times W = 14 \times 14.$$

conv4 branch.

$$\mathbf{M} = s(\mathbf{f}_4), \qquad H \times W = 28 \times 28.$$

conv4+conv5 multiscale branch. The two stages are fused first:

$$\tilde{\mathbf{f}}_5 = \mathrm{Up}(W_5 * \mathbf{f}_5)\,, \qquad \tilde{\mathbf{f}}_4 = W_4 * \mathbf{f}_4,$$

$$\mathbf{f}_{\text{fused}} = \mathrm{ReLU}\Big(\tilde{\mathbf{f}}_4 + \tilde{\mathbf{f}}_5\Big)\,,$$

then

$$\mathbf{M} = s(\mathbf{f}_{\text{fused}}), \qquad H \times W = 28 \times 28.$$

So the multiscale branch does *early fusion of features*, not separate losses on separate conv4 and conv5 maps.

1.4 Spatial Logit Pooling

The spatial maps are pooled into spatial concept logits:

$$\mathbf{c}_{\text{spatial}} = p(\mathbf{M}) \in \mathbb{R}^K,$$

where the residual coupling path uses LSE pooling in the current ablations.

1.5 Residual Concept Fusion

The final concept vector seen by the classifier is

$$\mathbf{c}_{\text{final}} = \mathbf{c}_{\text{global}} + \alpha\, \mathbf{c}_{\text{spatial}},$$

where α is the residual spatial coupling coefficient.

Important detail. In the current implementation, the image-level concept loss is applied to $\mathbf{c}_{\text{final}}$, not just $\mathbf{c}_{\text{global}}$. So the spatial branch affects training through the fused concept logits.

2 Loss Used in the 5-Run Ablation Study

The ablation study only used the following core terms:

$$\mathcal{L} = \lambda_{\text{global}}\mathcal{L}_{\text{global}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}} + \lambda_{\text{outside}}\mathcal{L}_{\text{outside}} + \lambda_{\text{coverage}}\mathcal{L}_{\text{coverage}} + \lambda_{\text{dice}}\mathcal{L}_{\text{dice}}.$$

In most runs:

$$\lambda_{\text{dice}} = 0.$$

Only the Dice ablation enabled the final term.

3 Loss Terms

3.1 1. Global Concept BCE on Fused Logits

Let $\mathbf{y} \in \{0,1\}^K$ denote the image-level concept targets. The global concept supervision is weighted BCE on the *fused* logits:

$$\mathcal{L}_{\text{global}} = \frac{\sum_{k=1}^{K} w_k^{(\text{global})}\ \mathrm{BCEWithLogits}(c_{\text{final},k}, y_k)}{\sum_{k=1}^{K} w_k^{(\text{global})}}.$$

Interpretation. This term trains the final concept vector used by the sparse classifier. Because $\mathbf{c}_{\text{final}}$ contains the residual spatial contribution, the spatial branch participates in image-level concept learning.

3.2 2. Soft-Align KL Loss on Spatial Maps

For a supervised concept k, let:

$$\mathbf{m}_k \in \mathbb{R}^{H' \times W'}$$

be the interpolated predicted concept map logits and

$$\mathbf{t}_k \in [0,1]^{H' \times W'}$$

be the target soft box / mask target.

The target is normalized into a spatial distribution:

$$\mathbf{q}_k = \frac{\max(\mathbf{t}_k, 0)}{\sum_u \max(t_{k,u}, 0)}.$$

The prediction becomes a log-distribution:

$$\log \mathbf{p}_k = \log \operatorname{softmax}(\mathbf{m}_k).$$

Then the local alignment loss is

$$\mathcal{L}_{\text{align}} = D_{\text{KL}}(\mathbf{q}_k \,\|\, \mathbf{p}_k).$$

Interpretation. This encourages the spatial map to put probability mass where the target region lies, without forcing hard binary pixel classification.

3.3 3. Outside-Mass Penalty

Let

$$\tilde{\mathbf{p}}_k = \operatorname{softmax}(\mathbf{m}_k)$$

over spatial locations. The outside-mass penalty is

$$\mathcal{L}_{\text{outside}} = \sum_u \tilde{p}_{k,u}\big(1 - \operatorname{clip}(t_{k,u}, 0, 1)\big).$$

Interpretation. This penalizes probability mass outside the target support and reduces spillover.

3.4 4. Coverage Penalty

Let

$$\hat{\mathbf{p}}_k = \sigma(\mathbf{m}_k).$$

Coverage over the target region is

$$\text{coverage}_k = \frac{\sum_u \hat{p}_{k,u} t_{k,u}}{\sum_u t_{k,u} + \varepsilon}.$$

The coverage loss is

$$\mathcal{L}_{\text{coverage}} = 1 - \text{coverage}_k\,.$$

Interpretation. This pushes the map to cover more of the target region rather than only peaking inside it.

3.5 5. Dice Loss

Only the Dice ablation used this term. With sigmoid probabilities:

$$\hat{\mathbf{p}}_k = \sigma(\mathbf{m}_k),$$

the Dice loss is

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2\sum_u \hat{p}_{k,u} t_{k,u} + \varepsilon}{\sum_u \hat{p}_{k,u} + \sum_u t_{k,u} + \varepsilon}.$$

Interpretation. Dice directly rewards overlap between predicted support and target support.

4 Ablation-Specific Loss Stacks

conv5 + cov025.

$$\mathcal{L} = \mathcal{L}_{\text{global}} + 0.25\,\mathcal{L}_{\text{align}} + 0.25\,\mathcal{L}_{\text{outside}} + 0.25\,\mathcal{L}_{\text{coverage}}.$$

conv5 + cov050. Same as above, but with

$$\lambda_{\text{coverage}} = 0.5.$$

conv5 + dice010. Same as ‘conv5 + cov025‘, plus

$$\lambda_{\text{dice}} = 0.1.$$

conv4. Same loss stack as ‘conv5 + cov025‘, but with the spatial branch operating on conv4 features.

conv4+conv5. Same loss stack as ‘conv5 + cov025‘, but with the spatial branch operating on the fused conv4+conv5 feature map.

5 Summary

The 5-run ablation study varied two things:

- the spatial feature source: conv5, conv4, or conv4+conv5,
- the local overlap regularization: baseline coverage, higher coverage, or Dice.

  The key design point is:

- concept supervision is applied to the fused logits $\mathbf{c}_{\text{final}}$,
- localization supervision is applied directly to the spatial maps $\mathbf{M}$.